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TECHNICAL NOTE

DETERMINING THE MOVEMENTS OF THE SKELETON USING WELL-CONFIGURED MARKERS

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Abstract—The problem of determining skeletal movements in three dimensions by using a number of landmarks is treated. We present a method that determines the motion of a rigid body by using the positions of the landmarks in least-squares sense. The method uses the singular value decomposition of a matrix derived from the positions of the landmarks. We show how one can use this method to express movement of skeleton segments relative to each other. As many others have pointed out, the movement can be very ill determined if the landmarks are badly configured. We present a condition number for the problem with good geometrical properties. The condition number depends on the configuration of the landmarks and indicates how to distribute the landmarks in a suitable way.

NOTATIONS

We will use some linear algebra conventions from scientific computing and technical mathematics in general. If the opposite is not explicitly stated, capital letters (A, Σ, etc.) denote matrices and small latin letters (a, b, etc.) denote column vectors. To measure the distances we use the *Euclidean vector norm* defined as

$$\|b\| = (b^{\mathrm{T}}b)^{1/2} = (b_1^2 + \cdots + b_n^2)^{1/2}, \tag{1}$$

where b is a real vector with n elements. For matrices we use a generalization of the Euclidean vector norm, generally called the *Frobenius norm* defined as

$$\|A\| = \left(\sum_{i,k} a_{ik}^2\right)^{1/2}, \tag{2}$$

where the sum is computed over all elements a_{ik} of the matrix A.

To get simple notations, we also define the set of 3×3 orthogonal rotation matrices as

$$\Omega = \{R \mid R^{\mathrm{T}}R = RR^{\mathrm{T}} = I_3;\ \det(R) = 1\}, \tag{3}$$

where R^{T} denotes the transpose of the matrix R.

INTRODUCTION

The problem of determining a rigid-body movement from noisy spatial coordinates of land-marks, anatomical or implanted, is treated by using a least-squares technique. (We will only treat the displacement as a movement from one discrete 'pose' to another, i.e. we are not discussing the continuous path of the displacement.) In Woltring (1991), different representations of the movement are discussed. We will use a three-dimensional translation vector d and a 3×3 orthogonal rotation/attitude matrix R determined by three parameters, as a representation.

To determine the rotation matrix R is a nonlinear problem. Older methods need initial guesses on the three parameters determining the matrix, and then they iteratively solve linearizations of the problem until convergence to a specified accuracy is reached. Spoor and Veldpaus (1980) describe an algorithm that needs no initial guesses and that has been recommended in other works (see e.g. Holtzreiter, 1991). Veldpaus *et al.* (1988) suggest slight improvements of this algorithm by which they avoid the computation of the eigenvectors to a 3×3 matrix. They also show that the accuracy of the rotation matrix depends on the distribution of the landmarks. We will present a method which refines the method described by Spoor and Veldpaus (1980). The method uses the singular value decomposition (see Golub and van Loan, 1989) of a 3×3 matrix C derived from the positions of the markers, instead of computing the eigenvalue decomposition of the more ill-conditioned matrix $C^{\mathrm{T}}C$. (The exact definition of the matrix C is given in Algorithm 1.) This change improves the stability and is essential for ill-conditioned problems. Since the method uses the singular value decomposition we will refer to the method as the SVD method.

The SVD method is derived from the *orthogonal Procrustes problem* which was treated by Schönemann (1966) and analyzed by Hanson and Norris (1981). The advantage with this method compared to those previously mentioned, besides the stability considerations, is that we get straightforward formulas which are easy to implement and fairly easy to analyze. Since the method has been published earlier (see e.g. Arun *et al.*, 1987; Hanson and Norris, 1981), we will only present it without any derivation. Instead, we will focus on how the method can be used for analyzing relative movements and to describe the stability of the problem.

There are many papers written about how errors in the positions of the landmarks influence the determination of the movement. For example, Woltring *et al.* (1985) and De Lange *et al.* (1990) study how the helical axis characteristics are affected. We have taken the ideas of Veldpaus *et al.* (1988), who showed that the distribution of the landmarks is very critical, a bit further. We suggest that a condition number for the configuration of landmarks should be used. The condition number, based on singular values, tells how the measurement errors in the positions of the landmarks affect the computed rotation matrix. We will also show that the condition number has a clear geometrical interpretation that tells us how far the landmarks are distributed from a straight line.

The methods that are described in this paper have been used at the University Hospital in Umeå Sweden, in a program system developed from the old RSA system presented

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by Selvik (1989). For references to applications we suggest Kärrholm (1989) and the references cited therein.

THE PROCRUSTES PROBLEM FOR RIGID-BODY MOVEMENT

Assume that we have n landmarks in a rigid body and let $\{x_1, \ldots, x_n\}$ be the three-dimensional positions of these landmarks before the movement and $\{y_1, \ldots, y_n\}$ be the positions after the movement. We want to determine a rotation matrix R and a translation vector d that map the points x_i to the points y_i, $i=1, \ldots, n$. Because of measurement errors the mapping is not exact and we use the following least-squares problem:

$$\min_{R\in\Omega, d} \sum_{i=1}^{n} \|Rx_i+d-y_i\|^2 \tag{4}$$

to determine R and d. Problem (4) is linear with respect to the vector d but it is nonlinear with respect to the rotation matrix R (that is because of the orthogonality condition of R). In Arun *et al.* (1987) and Hanson and Norris (1981), it is shown that we can use the *singular value decomposition* of the matrix $C=BA^{\mathrm{T}}$ to compute the solution to (4). The algorithm below describes the computation steps.

Algorithm 1. *Algorithm for solving problem* (4) *by using the SVD method.*

(1) $\bar{x}=\frac{1}{n}\sum_{i=1}^{n} x_i, \bar{y}=\frac{1}{n}\sum_{i=1}^{n} y_i$,

(2) $A=[x_1-\bar{x}, \ldots, x_n-\bar{x}]$, $B=[y_1-\bar{y}, \ldots, y_n-\bar{y}]$,

(3) $C=BA^{\mathrm{T}}$,

(4) $P\Gamma Q^{\mathrm{T}}=C$. (*Computation of the singular decomposition of C, i.e. P and Q are 3×3 orthogonal matrices and* $\Gamma=\mathrm{diag}(\sigma_1(C), \sigma_2(C), \sigma_3(C))$. *The numbers $\sigma_i(C)$ are the singular values of the matrix C and satisfy* $\sigma_1(C)\geqslant\sigma_2(C)\geqslant\sigma_3(C)\geqslant 0$.)

(5) $R=P\,\mathrm{diag}(1, 1, \det(PQ^{\mathrm{T}}))Q^{\mathrm{T}}$,

(6) $d=\bar{y}-R\bar{x}$.

The solution to problem (4) is unique if $\sigma_2(C)$ is nonzero. (More strictly, the solution is not unique if $\sigma_2(C)$ equals zero or if $\sigma_2(C)=\sigma_3(C)$ and $\det(PQ^{\mathrm{T}})=-1$. But the latter situation implies that a reflection is the best othogonal matrix describing the movement. This situation will seldom occur when studying rigid bodies and indicates that something is wrong.)

The movement computed by Algorithm 1 shall be interpreted as a rotation around a *base point bp* equal to the origin, followed by the translation d. It may sometimes be desirable to use another base point. The rotation matrix R is invariant for the choice of base point, but the translation vector is not. If one wants to use another base point bp, it is only needed to replace step 6 in the algorithm by

$$d=\bar{y}-bp-R(\bar{x}-bp). \tag{5}$$

THE RELATIVE MOVEMENT OF A SEGMENT WITH RESPECT TO A REFERENCE SEGMENT

We wish to compute the *relative movement* of a skeletal segment, here called *actual segment*, with respect to a *reference segment*.

The motion to be computed is assumed to start at *start time* and end at *end time*. In order to compute a relative movement, we have to solve two approximation problems of the kind defined by problem (4). First we have to solve one approximation problem to derive the *absolute motion* of the reference segment between the two times. That has to be done because we do not want the movement of the whole object, i.e. both the reference segment and the actual segment, to affect the relative movement (we only want to determine the movement of the actual segment with respect to a 'fixed' reference segment). We solve problem (4) with $\{x_1, \ldots, x_n\}$ being the position at start time of n landmarks in the reference segment and $\{y_1, \ldots, y_n\}$ the position of the landmarks at end time. In this way we derive the orthogonal rotation matrix R_{Ar} and translation vector d_{Ar}. The subindex Ar is intended to give the information that R_{Ar} and d_{Ar} define the *absolute* movement of the *reference* segment between the start time and end time.

Let $\{u_1, \ldots, u_{n'}\}$ be the positions of n' landmarks in the actual segment at start time that have been moved into the positions $\{v_1, \ldots, v_{n'}\}$ at the end time. (Note that the number of landmarks n' in the actual segment does not have to be the same as the number of landmarks n in the reference segment.) To be able to determine the relative movement we use the fact that the transformation

$$y=R_{\mathrm{Ar}}x+d_{\mathrm{Ar}} \tag{6}$$

determines the absolute movement of the reference segment. Thus, R_{Ar} and d_{Ar} can be used to transform the landmarks in the actual segments from the positions $\{v_1, \ldots, v_{n'}\}$ to the positions $\{v'_1, \ldots, v_{n'}\}$ which they would have if they were not affected by the absolute movement of the reference segment. The transformation is

$$v'_i=R_{\mathrm{Ar}}^{\mathrm{T}}v_i-R_{\mathrm{Ar}}^{\mathrm{T}}d_{\mathrm{Ar}}, \quad i=1, \ldots, n'. \tag{7}$$

The relative movement can then be computed as the solution to another approximation problem of the kind defined by problem (4), such that

$$v'_i\approx R_{\mathrm{R}}u_i+d_{\mathrm{R}}, \quad i=1, \ldots, n'. \tag{8}$$

Of course, the subindex R here stands for relative movement. This method of computing the relative movement is illustrated by Fig. 1.

SENSITIVITY ANALYSIS OF THE RIGID-BODY MOVEMENT

In this section we will show how the landmarks should be configured in the segment to give as much information about the movement as possible.

Assume that

$$y_i=Rx_i+d, \quad i=1, \ldots, n, \tag{9}$$

holds for a given rotation matrix R and translation vector d. We will now study how the rotation matrix R changes due to errors in the positions $\{x_1, \ldots, x_n\}$ and $\{y_1, \ldots, y_n\}$ of the landmarks.

If we insert the expressions for A and B, given in Algorithm 1, into problem (4), we can conclude that the rotation matrix is computed by solving

$$\min_{R\in\Omega} \|RA-B\|. \tag{10}$$

According to this problem the perturbed problem of determining the rotation matrix becomes

$$\min_{(R+\Delta R)\in\Omega} \|(R+\Delta R)(A+\Delta A)-(B+\Delta B)\|, \tag{11}$$

where $\Delta A=[\delta x_1-\overline{\delta x}, \ldots, \delta x_n-\overline{\delta x}]$, $\Delta B=[\delta y_1-\overline{\delta y}, \ldots, \delta y_n-\overline{\delta y}]$.

The following theorem, which is proved in Söderkvist (1992) shows how the rotation matrix R is asymptotically affected by perturbations $\delta x_i, \delta y_i$, $i=1, \ldots, n$.

Theorem 1. *Assume that*

- $B=RA$, $A, B\in R^{3\times n}$, $R\in\Omega$,
- *A and B have the singular values $\sigma_1\geqslant\sigma_2\geqslant\sigma_3\geqslant 0$, where $\sigma_2>0$.*

Let the orthogonal matrix $R+\Delta R$ be the solution to the perturbed problem

$$\min_{(R+\Delta R)\in\Omega} \|(R+\Delta R)(A+\Delta A)-(B+\Delta B)\|. \tag{12}$$

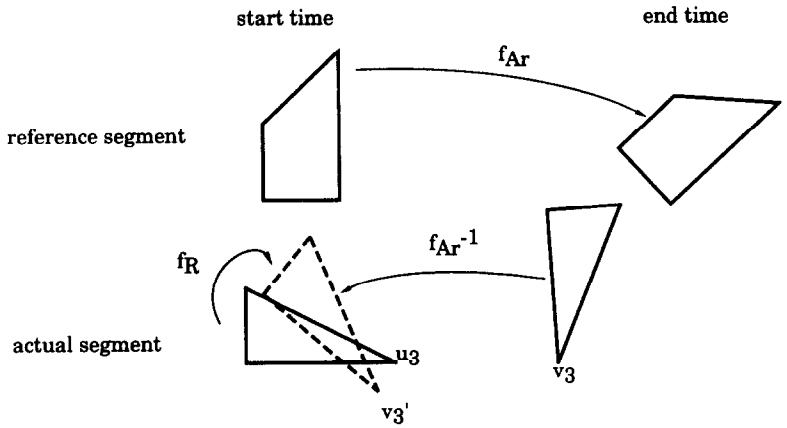

Fig. 1. The computation of relative movement. The absolute movement $f_{Ar}(x_i) = R_{Ar}x_i + d_{Ar}$ of the points x_i in the reference segment is computed. The inverse of this movement is then used to transform the points in the actual segment from the positions v_i to the positions $v_i' = R_{Ar}^T v_i - R_{Ar}^T d_{Ar}$. At last, the relative movement $f_R(u_i) = R_R u_i + d_R$ is computed such that $f_R(u_i) \approx v_i'$.

A first-order bound of $\|\Delta R\|$ *is given by*

$$\lim_{\substack{\varepsilon_A \to 0 \\ \varepsilon_B \to 0}} \sup_{\substack{\|\Delta A\| \leq \varepsilon_A \\ \|\Delta B\| \leq \varepsilon_B}} \frac{\|\Delta R\|}{\|\Delta A\| + \|\Delta B\|} = \frac{\sqrt{2}}{(\sigma_2^2 + \sigma_3^2)^{1/2}}. \tag{13}$$

The theorem shows that the problem of determining the rotation matrix is very ill conditioned when the two smallest singular values σ_2 and σ_3 of A are small, that is, when the landmarks are almost on a straight line.

Using Theorem 1, we define the condition number for the problem of determining the rigid-body movement as

$$\kappa_A = \frac{1}{(\sigma_2^2 + \sigma_3^2)^{1/2}}. \tag{14}$$

The number κ_A is a condition number of the configuration with respect to an absolute error in the configuration. If we consider relative error in the configuration instead, the condition number can be defined as

$$\kappa_R = \frac{\|A\|}{(\sigma_2^2 + \sigma_3^2)^{1/2}}. \tag{15}$$

The relative condition number κ_R also indicates how well a segment with bounded size has been used for the distribution of the landmarks. The absolute condition number κ_A, on the other hand, often gives the best measure, $\sqrt{2}\kappa_A(\|\Delta A\| + \|\Delta B\|)$, of the worst error in the rotation matrix. This is the case because the absolute accuracy in the positions of the landmarks are often the same regardless of the size of the segment.

Since

$$\|A\| = (\sigma_1^2 + \sigma_2^2 + \sigma_3^2)^{1/2}$$

holds, the condition number κ_R is always greater than or equal to $\sqrt{3/2}$. The equality occurs when all the singular values are equal, i.e. when the landmarks $\{x_1, \ldots, x_n\}$ are isotropically distributed, as in an equilateral tetrahedron, a cube, etc. For a two-dimensional configuration κ_R is always greater or equal to $\sqrt{2}$.

Since the translation vector d is computed as step 6 in Algorithm 1, it follows that the perturbed problem has the solution

$$d + \delta d = \overline{y + \delta y} - (R + \Delta R)(\overline{x + \delta x}). \tag{16}$$

From the triangular inequality we get the following bound for the resulting perturbation δd:

$$\|\delta d\| \leq \|\overline{\delta y}\| + \|\overline{\delta x}\| + \|\Delta R\|(\|\bar{x}\| + \|\overline{\delta x}\|). \tag{17}$$

If we assume that the errors $\{\delta x_1, \ldots, \delta x_n\}$ and $\{\delta y_1, \ldots, \delta y_n\}$ are distributed such that their mean values become small, we conclude that the translation vector is well determined if $\bar{x}$ is close to the origin and if the problem of determining the rotation matrix is well conditioned.

GEOMETRICAL ASPECTS OF THE CONDITION NUMBER

A necessary condition for the approximation problem (4) to have a unique solution is that the matrix BA^T has at least rank equal to 2. Hence, if the landmarks $\{x_1, \ldots, x_n\}$ are on a straight line we do not have enough information to derive the rotation matrix R and translation vector d describing the movement. This is also indicated by Theorem 1. Thus, to get an idea of how well configured the set of landmarks $\{x_1, \ldots, x_n\}$ is, it is natural to try to find out how close these landmarks are to a straight line. For this purpose we use the singular value decomposition $A = W[\Sigma, 0]V^T$ of the matrix $A = [x_1 - \bar{x}, \ldots, x_n - \bar{x}]$ with the singular values σ_1, σ_2, and σ_3. In Söderkvist and Wedin (1991), it is shown that this decomposition gives us the following geometrical properties about the distribution of the landmarks (see Fig. 2):

- The square sum of the distances of the landmarks to the 'closest' straight line equals $\sigma_2^2 + \sigma_3^2$.
- The closest straight line passes through $\bar{x}$ and has the direction given by the first column in W.
- The square sum of the distances of the landmarks to the closest plane equals σ_3^2.
- The closest plane passes through $\bar{x}$ and an orthogonal base for the plane is given by the two first columns in W.

Thus, the condition number κ_A has the clear geometrical interpretation as the square root of the inverse of the square sum of the distances of the landmarks to a straight line. Therefore one should strive to put the landmarks as widely separated as possible and avoid putting them close to a straight line.

From the statements above we see that the two first columns in W give the 'leading and second leading direction of orientation' of the landmarks. We can also say how much additional information an extra landmark with position x_{n+1} will give. If the position x_{n+1} is close to the closest line

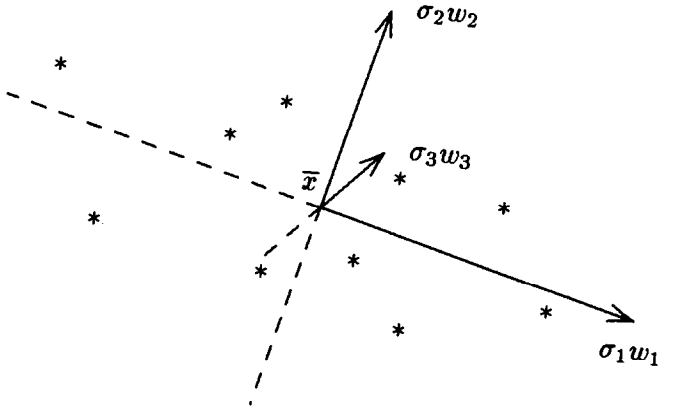

Fig. 2. A three-dimensional illustration of the geometrical distribution of the landmarks (*) is given by the singular value decomposition of the matrix A. The orthogonal vectors w_1, w_2, and w_3 are the columns of the matrix W.

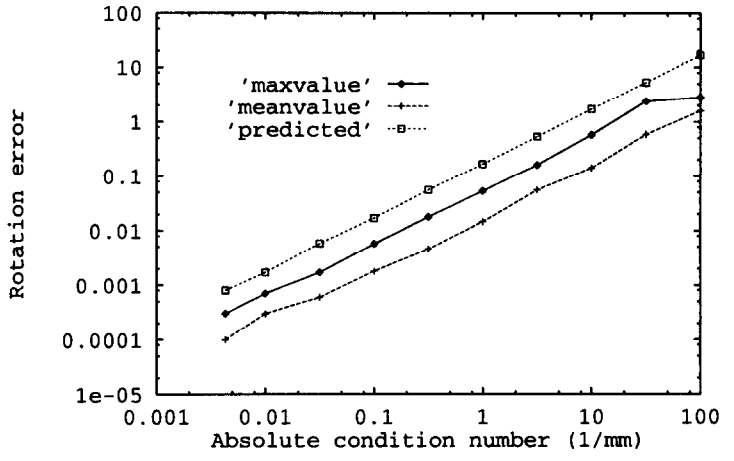

Fig. 3. The meanvalue, maximal value, and predicted maximal value of $\|R-R_i\|$ are plotted against the absolute condition number k_A.

we do not include much more information and the condition number κ_A only decreases by a small amount. (The condition number κ_A can never increase by using an extra landmark.) On the other hand, if x_{n+1} is located far away from the 'closest' line we get a useful decrease of the condition number and the problem of determining the rigid-body movement becomes better conditioned.

From the discussion above, we propose that different landmarks give a different amount of information about the movement. This also implies that good precision of the measurements of the positions of the landmarks far from the closest line is more important than of the landmarks close to the closest line.

NUMERICAL EXPERIMENTS

In order to investigate the validity of the theoretically derived condition numbers, we have performed dummy motion experiments where normally distributed random perturbations have been put into the positions of the landmarks. For each of 10 different configuration of landmarks we have computed 100 different rotation matrices R_i, where each R_i is computed from perturbed positions of landmarks. The perturbations in the positions of the landmarks are sampled from an unbiased normal distribution with standard deviation equal to 10 μm.

These rotation matrices are compared to a fixed matrix R computed from unperturbed positions of landmarks. The positions of the landmarks are chosen such that the movement is exactly described by a rotation matrix R, that corresponds to three Cardan angles all equal to 5°. For each configuration we describe the errors in the rotation matrix by the following three quantities:

$$\text{meanvalue} = \frac{1}{100}\sum_{i=1}^{100} \|R_i - R\|,$$

$$\text{maxvalue} = \max_{1 \leq i \leq 100} \|R_i - R\|,$$

$$\text{predicted} = \kappa_A \sqrt{2}\varepsilon,$$

where ε is the largest sum $\|\Delta A\| + \|\Delta B\|$ of the 100 sampled sets of perturbations. (Since all of the matrices that represent the different sets of perturbations approximately have the same norm, $\|\Delta A\| + \|\Delta B\| \approx 100$ μm, this choice is not critical.) The quantity *predicted* is a first-order estimate of the worst possible error obtained from Theorem 1.

We have used 10 different configurations of landmarks. First the landmarks are the vertices in a perfectly configured three-dimensional cube with center at the origin and with the

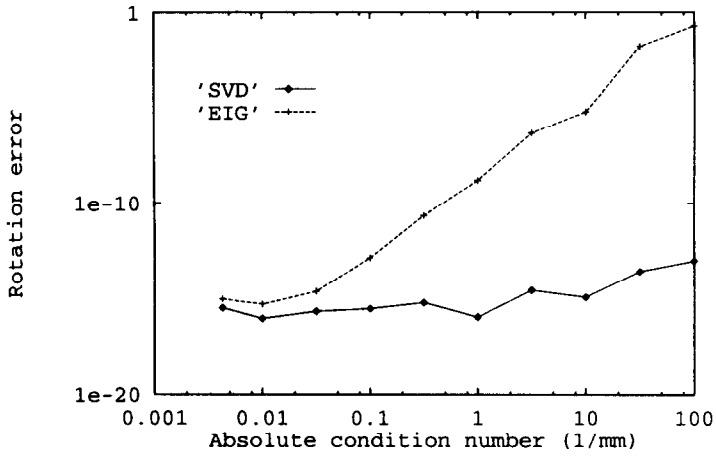

Fig. 4. The errors in the computed rotation matrices corresponding to the SVD and the Eigenvalue method, respectively, are plotted against the absolute condition number k_A.

volume 125 cm^3. For each of the following configurations the landmarks are put closer to the x-axis and form different rectangular boxes. The condition numbers increase since the landmarks approach a straight line.

As one can see in Fig. 3, the errors in the rotation matrix increase with increasing condition number. The first-order estimate of the worst error, *predicted*, seems to be useful for perturbations of this small size.

To compare the stability of the SVD method with the Eigenvalue method described by Spoor and Veldpaus (1980), we have computed the rotation matrices R_{SVD} and R_{EIG} for different configurations by using a machine precision of 10^{-16}. The errors introduced in the computations are measured by taking the norm of the difference of the computed matrices to an exact rotation matrix R, i.e. $\|R_{SVD}-R\|$ and $\|R_{EIG}-R\|$. The result shown in Fig. 4 illustrates that the SVD method is stable and hence superior for ill-conditioned problems.

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